Compressing the Latent Communication Channel of a Recursive Multi-Agent System: Accuracy, Churn, and Trajectory Fidelity

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Abstract

Latent-space multi-agent systems exchange dense activation vectors rather than text. We study whether these messages can be quantized aggressively without materially changing system behavior. We insert the data-oblivious, mean-squared error (MSE) core of TurboQuant into every latent link of the released RecursiveMAS system, without changing model weights or prompts. Our primary study runs locally on one RTX 5070 Ti and covers five task/tier cells: the 2×2 of {Math500, MBPP+} $\times$ {light, scaled} plus MedQA with the light tier. None of the five sampled ladders from 8 to 2 bits per coordinate ($4\times$–$16\times$ nominal compression) exhibits a monotonic rate–accuracy decline. In the four non-confounded paired-greedy cells, accuracy differences are $+2.0$, -2.4, 0, and -2.0 percentage points with confidence intervals spanning zero, while 4.4–10.0% of individual answers change. Corrected top-K capture analysis further shows that 51.2–92.8% of primary greedy sequences diverge within a 128-position window. On the MBPP+ task the scaled tier is markedly more trajectory-robust than the light tier (51.2% versus 92.8% divergence; matched-prefix KL 0.059 versus 0.113 nats), despite nearly identical measured channel cosine. Completing the 2×2 shows this contrast is *task-specific*: on Math500 the same light-to-scaled change barely moves divergence (86.4% versus 80.4%). It is therefore an exploratory, task-conditioned tier association, not a causal model-capacity law. Our evidence supports aggregate answer robustness under aggressive fake quantization, but not bit-exact losslessness, trajectory preservation, or a deployable bandwidth claim. An earlier independent Kaggle T4 Math500 ladder shows the same aggregate pattern under fp32.

1 Introduction

Multi-agent language-model systems increasingly communicate through internal activation vectors. Such latent messages avoid decoding and re-encoding text, but they can contain thousands of floating-point coordinates and may be sent many times during recursive reasoning. In a distributed deployment, the communication channel can therefore become a systems bottleneck.

Quantization is well studied for model weights and the attention KV cache. A latent inter-agent message is a different target: it is consumed once, transformed by a learned cross-model adapter, and repeatedly replaced as agents refine an answer. Small local errors may be harmless, may accumulate, or may push greedy decoding across token decision boundaries. These possibilities cannot be resolved from single-vector reconstruction error alone.

We study RecursiveMAS [Yang et al., 2026], a released recursive latent multi-agent system, and the data-oblivious MSE core of TurboQuant [Zandieh et al., 2026]. The intervention randomly rotates and scalar-quantizes every inter-agent vector. We measure three distinct outcomes: channel reconstruction, final-answer behavior, and token-trajectory fidelity. Separating them matters. A system can preserve aggregate benchmark accuracy while changing which individual problems it solves and which reasoning text it generates.

Our contributions are:

- an instrumentation-only implementation that quantizes all RecursiveMAS latent links while preserving the released models and evaluation pipeline;
- a primary five-cell local study (the 2×2 of {Math500, MBPP+} $\times$ {light, scaled} plus MedQA) with paired answer churn rather than net accuracy alone;
- sampled bit-rate ladders with no detected monotonic aggregate-accuracy degradation at $4\times$–$16\times$ nominal payload compression;
- a corrected top-K trajectory analysis that excludes conditional retry calls and avoids double-counting probability mass in the residual tail;
- an exploratory same-task result in which the scaled MBPP+ system is much more trajectory-robust than the light system, shown to be *task-specific* because it does not reproduce on Math500; and
- documentation of two measurement failures: pre-Ampere bf16 numerical collapse and a greedy multiple-choice option bias.

This is an empirical study, not a new compression algorithm. We evaluate the MSE core of TurboQuant and omit its QJL inner-product residual.

2 System and intervention

RecursiveMAS. RecursiveMAS connects heterogeneous language-model agents through learned adapters and iterates latent messages across planning, solving, and critique stages. We pin the upstream source at commit `f95d512`. The completed matrix uses `sequential_light` and two `sequential_scaled` contrasts (MBPP+ and Math500). The upstream checkout is treated as read-only: the runner verifies its commit, creates a disposable source copy in the run directory, and instruments only that copy.

Quantizer. For $x \in \mathbb{R}^d$, TurboQuant-MSE normalizes x, applies a fixed Haar-random orthogonal transform, quantizes each transformed coordinate with a 2^b-level Lloyd–Max codebook for the concentrated marginal, applies the inverse transform, and restores the norm. We report

$$E_{\mathrm{rel}} = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2}$$

and cosine similarity at every wrapped adapter call. Quantization arithmetic is performed in fp32 and returned in the pipeline dtype.

Scope of compression. The ratios $32/b$ are nominal packed-payload ratios. Our experiments use fake quantization: the receiver consumes the same reconstructed vector that a low-bit transport would deliver, but we do not serialize codes or measure network latency. Metadata, rotation cost, and codec overhead are therefore outside the stated ratio.

3 Experimental design

3.1 Conditions

The primary study uses one RTX 5070 Ti 16 GB under WSL2, native bf16, seed 42, three recursive rounds, and $n = 250$ in every cell. The matrix contains:

- `sequential_light` on Math500, MBPP+, and MedQA;
- `sequential_scaled` on MBPP+ and Math500.

Earlier independent runs used Kaggle T4 fp32 for a sampled Math500 ladder and Modal A100 fp32 for controls and a depth sweep. They corroborate the broad Math500 result and expose dtype hazards, but are not the primary reproduction target.

Sampled ladders use the released sampling configuration at bits $\{0, 8, 4, 2\}$. Paired fidelity uses greedy decoding for REF ($b = 0$) and INT4 ($b = 4$), identical input order, and identical seed.

3.2 Outcome levels

Channel fidelity. We record cosine similarity, relative L2 error, norm ratio, and call counts.

Answer behavior. We report accuracy, paired delta, discordant counts, exact McNemar tests, bootstrap intervals, and churn: the fraction of samples whose correctness indicator changes. A two-one-sided equivalence test uses a pre-specified ± 2 percentage-point margin, but failure to establish equivalence is reported as inconclusive.

Trajectory behavior. Generation capture stores top-K logits and residual tail log-mass for up to 128 positions locally ($K = 256$). Only the fixed primary generate calls—exactly $\lceil n/\text{batch size} \rceil$ calls—are paired. Conditional answer-retry calls are excluded because REF and INT4 need not trigger them on the same examples. At each matched position, probabilities are reconstructed on the union of the two top-K supports. Mass assigned to newly exposed union tokens is subtracted from the residual tail before normalization, preventing double counting.

We report whether the greedy sequences diverge within the capture window, their common-prefix length, and KL/JS on the matched prefix. This KL remains approximate and selection-conditioned: once tokens diverge, the next-token contexts differ.

4 Results

4.1 The real channel matches the expected distortion rate

Table 1 compares the real Solver adapter with synthetic Gaussian-on-sphere vectors. The 4-bit real-channel error, $E_{\text{rel}} = 0.0093$, matches the expected TurboQuant rate and has cosine about 0.995.

4.2 Five-cell local answer behavior

Table 2 extends the accuracy observation across both tasks and both tier contrasts. No sampled ladder is monotonically ordered by rate. In the four non-confounded cells (Math500/light, Math500/scaled, MBPP+/light, MBPP+/scaled), the paired greedy deltas are small and their

Table 1: Relative squared reconstruction error of TurboQuant-MSE.

bits	synthetic ($d = 2048$)	real adapter ($d = 1536$)	reported rate
8	0.0001	0.0001	approximately 0
4	0.0095	0.0093	0.009
3	0.0345	0.0339	0.030
2	0.1175	0.1159	0.117

Table 2: Local $n = 250$ answer results. Ladder order is REF/8/4/2 bits. Churn is paired greedy correctness churn.

cell	sampled ladder (%)	greedy REF/INT4 (%)	delta, 95% CI	churn
Math500 / light	77.6/76.0/78.8/78.0	76.8/78.8	+2.0 $[-2.0, +6.0]$	10.0%
Math500 / scaled	82.8/85.6/84.0/84.8	87.6/85.2	-2.4 $[-6.0, +1.2]$	8.8%
MBPP+ / light	32.8/33.6/38.8/34.8	36.4/36.4	0.0 $[-4.0, +4.0]$	9.6%
MBPP+ / scaled	70.8/73.6/72.0/71.2	74.4/72.4	-2.0 $[-4.8, +0.4]$	4.4%
MedQA / light	34.4/26.4/32.8/30.0	21.2/36.4	+15.2 $[+8.8, +21.6]$	30.4%

95% intervals span zero. Nevertheless, 4.4–10.0% of correctness labels change. Thus a near-zero net delta can conceal offsetting losses and gains.

The MedQA greedy row is confounded. REF selects option A about 46% of the time, whereas the sampled REF has no comparable bias. INT4 dither disrupts the bias and appears to improve accuracy. Because the reference condition is pathological, this row cannot estimate channel fidelity; the non-monotonic sampled ladder is the valid task-level observation.

4.3 Independent cloud Math500 corroboration

An earlier fp32 T4 ladder at $n = 250$ is flat within its uncertainty (Table 3, Figure 1). At 2 bits, REF and the quantized system both solve 188/250 problems. These unpaired stochastic runs support "no detected degradation," not exact equality, and provide an independent hardware/dtype corroboration of the local Math500 pattern.

4.4 Corrected trajectory fidelity

Table 4 reports the corrected primary-call analysis. Individual channel cosine is almost identical in all cells, as expected from a data-oblivious quantizer at fixed rate. Downstream consequences are not identical. The light cells diverge in 86.4–96.4% of primary sequences within 128 positions. Scaled MBPP+ diverges in only 51.2%, retains a longer common prefix, and has lower matched-prefix KL—but scaled Math500 diverges in 80.4%, close to its light counterpart (86.4%), so the tier effect is large on MBPP+ and small on Math500.

The cleanest comparison holds task and local environment fixed: the light and scaled MBPP+ tiers, where scaling nearly halves divergence. Critically, this contrast does *not* reproduce on Math500, where the same light-to-scaled change moves divergence only 86.4% to 80.4% and even raises matched-prefix KL. A general capacity law would predict a comparable drop on both tasks; its absence on Math500 means the effect is task-specific. The result is compatible with larger or more capable models absorbing channel noise more effectively *on some tasks*, but capacity is not isolated: the tier also changes architecture, checkpoint family, baseline competence, margins, and generated length. Equal cosine likewise describes geometric reconstruction, not equal semantic

Table 3: Independent historical Math500 sampled ladder (T4 fp32, $n = 250$).

bits	nominal compression	accuracy	delta versus REF
REF	1×	75.2%	—
8	4×	78.4%	+3.2 pp
4	8×	76.8%	+1.6 pp
2	16×	75.2%	0.0 pp

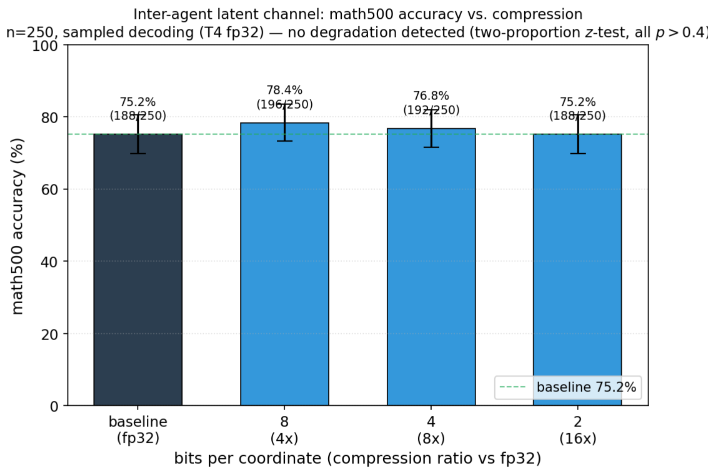

Figure 1: Independent T4 Math500 sampled accuracy versus bit rate. Error bars overlap; the experiment detects no monotonic degradation.

perturbations. We therefore call this a task-conditioned tier association and treat a causal capacity claim as a hypothesis for replication.

5 Measurement failures and reproducibility lessons

Hardware precision. On pre-Ampere GPUs, the released bf16 configuration silently collapses RecursiveMAS recursive inference to roughly 30–35% Math500 accuracy. Forcing fp32 restores the baseline. A separate A100 run initially showed a 19-point quantized drop because repeated bf16–fp32 boundary casts were introduced by the instrumentation. A dtype-coherent pipeline removes the artifact. These are pipeline failures, not compression results.

Conditional generation calls. The first Tier-2 local analysis paired every captured `generate()` call. Answer extraction can trigger a retry conditionally, so after REF and INT4 diverge those calls are not structurally aligned. Restricting analysis to the fixed primary calls removes this bias.

Top-K residual mass. When a token appears in only one captured top-K, its approximated probability in the other distribution must be deducted from that distribution's residual tail. Failing to do so double counts mass and inflates KL. The corrected analyzer and synthetic regression tests implement this accounting. Historical cloud KL values produced by the legacy estimator are retained for auditability but are not compared numerically with Table 4.

Table 4: Corrected local Tier-2 fidelity. Divergence is windowed at 128 captured positions; KL is approximate and restricted to matched prefixes.

cell	diverged	common prefix / 128	KL (nats)	channel cosine
Math500 / light	86.4%	53.7	0.079	0.9952
Math500 / scaled	80.4%	54.6	0.147	0.9953
MBPP+ / light	92.8%	35.8	0.113	0.9953
MBPP+ / scaled	51.2%	65.7	0.059	0.9953
MedQA / light	96.4%	32.4	0.045	0.9952

6 Discussion

The central result is a dissociation. Aggressive channel quantization produces nearly constant aggregate accuracy at the resolution of these experiments, yet it changes individual answers and most light-tier greedy trajectories. "Answer preserving" is therefore meaningful only as a population-level shorthand; it must not be read as preserving every answer or every reasoning trace.

The MBPP+ result weakens a simple redundancy explanation. Code tokens are themselves the evaluated output, yet the light and scaled sampled ladders remain flat. The scaled tier's reduced churn and trajectory divergence on MBPP+ suggest that system competence or token margins may mediate robustness—but because the same tier change leaves Math500 divergence almost unchanged, any such mediator must itself be task-dependent. Teacher-forced analysis is needed to distinguish stable next-token distributions from the selection effect created by matched prefixes.

The nominal compression ratios are promising but incomplete. A real system must encode codebook indices, norms, seeds or transform identifiers, and packet metadata; it must also pay rotation and codec latency. The present work establishes behavioral fidelity of reconstructed messages, not end-to-end throughput.

7 Limitations and next steps

All local cells use one seed and one machine. The {math, code} $\times$ {light, scaled} 2×2 is now crossed, but only at a single seed; MedQA greedy is confounded, and MBPP+ contains only 378 available problems. Confidence intervals still allow effects of several percentage points. The trajectory window is 128 positions, local capture uses $K = 256$, and matched-prefix KL is not a teacher-forced causal comparison. We evaluate fake quantization and the TurboQuant MSE core only.

The highest-value next steps are: (i) repeat both MBPP+ tiers at three to five seeds (scaled Math500 is complete and shows the tier effect is task-specific); (ii) force REF tokens in both conditions and measure aligned KL, rank, and top-1 margins; (iii) sweep 2/3/4/6/8 bits with an unrotated scalar baseline; (iv) implement the QJL residual; (v) test a tool-sensitive deliberation topology and a high-baseline non-math task; and (vi) measure serialized bytes, latency, wall-clock time, and VRAM. Confirmatory claims should pre-specify the primary estimand, equivalence margin, seed pooling, and multiple-comparison policy.

8 Conclusion

The latent channel of RecursiveMAS tolerates aggressive data-oblivious fake quantization surprisingly well in aggregate accuracy across the completed cells. That robustness coexists with real behavioral change: individual answers churn and greedy trajectories often diverge. The scaled MBPP+ tier is substantially more trajectory-robust than the light tier, but completing the 2×2 shows this is task-specific—it does not reproduce on Math500—so it is a result worth testing causally, not yet a capacity law. The practical conclusion is precise: low-bit latent communication is a credible systems direction, but claims of losslessness require stronger statistics, position-aligned fidelity, broader tasks and architectures, and real transport measurements.

References

Xiyuan Yang, Jiaru Zou, Rui Pan, Ruizhong Qiu, Pan Lu, Shizhe Diao, Jindong Jiang, Hanghang Tong, Tong Zhang, Markus J. Buehler, Jingrui He, and James Zou. Recursive Multi-Agent Systems. *arXiv preprint arXiv:2604.25917*, 2026.

Amir Zandieh, Majid Daliri, Majid Hadian, and Vahab Mirrokni. TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2504.19874.